



Ref : LW/JPR/2026/1285

Date : 31st July 2026**TO WHOMSOEVER IT MAY CONCERN**

This is to certify that the project work and report entitled during the Summer Internship on **“Unlocking the Full Potential of Data Science, Machine Learning & Artificial Intelligence for Predictive Analytics, Intelligent Decision-Making, and AI-Driven Innovation – Unleashing the Power of Data Science and Artificial Intelligence for Business Growth and Future-Ready Solutions.”** embodies the original work of **Mr. Asim Azhar** from **IILM University, Greater Noida** at LinuxWorld Informatics Pvt Ltd.

The duration of the project undergone as mentioned above, under the mentorship of Mr. Syed Jibbran Ali, Sr. Technical Consultant was from 27th June 2026 – 31st July 2026.

Project Description:

The AI-Based Hiring Prediction System is a machine learning application designed to help recruiters decide if a candidate should be hired based on professional and academic details. The system looks at information like skills, certifications, education, work experience, number of projects, salary needs and AI test results to make hiring predictions.

The project starts with preparing the data, where unimportant columns are taken out and missing information is. Categories are changed into numbers. Text parts like skills, certifications and job roles are handled using TF-IDF to turn text into numbers.

The numbers and text parts are put together into one dataset. Then numerical data is adjusted using Standard Scaler to make the model work better. Many machine learning methods like Logistic Regression, Random Forest Support Vector Machine (SVM) and K-Nearest Neighbors (KNN) are used. These models are. Tested. Their performance is checked with accuracy and other measures to find the one.

A function is also added so recruiters can enter candidate details and get a hiring decision along with the chance of being chosen. This helps make the part of hiring faster and more consistent while saving time.

Technologies Used:

The AI-Based Hiring Prediction System was built using Python in Google Colab/Jupyter Notebook. The project uses Pandas and NumPy for handling and preparing data. Matplotlib and Seaborn are used to make charts and understand the data better. Machine learning models are created using Scikit-learn, including Logistic Regression, Random Forest Classifier, Support Vector Machine (SVM) and K-Nearest Neighbors (KNN). Text parts like skills, certifications and job roles are changed into numbers using TF-IDF. StandardScaler is used for adjusting numbers and Pipeline along with GridSearchCV helps in making the model. These tools helped in preparing data, training models testing them and making predictions.

● Address

Plot no 5, Krishna Tower, GopalNagar - A,
Gopalpura Bypass, Jaipur - 302015

● Phone

+91 93510 09002
0141 250 1609

● Web/Email

www.linuxworldindia.org
lr@lwnindia.com



Ref : LW/JPR/2026/1285

Date : 31st July 2026**Working Methodology:**

- Load the hiring dataset.
- Perform data exploration and visualization.
- Remove unnecessary columns.
- Encode categorical variables.
- Combine text fields (Skills, Certifications, Job Role).
- Convert text into TF-IDF vectors.
- Scale numerical features.
- Split the dataset into training and testing sets.
- Train multiple machine learning models.
- Compare model performances.
- Optimize the best model using GridSearchCV.
- Predict hiring decisions for new candidates.

Conclusion:

The AI-Based Hiring Prediction System shows how machine learning can help recruiters make more consistent hiring choices. By combining education, skills, certifications and experience with text processing like TF-IDF the system helps predict who gets hired.

The project makes it easier to screen candidates, makes decision-making faster and shows how machine learning can be used in Human Resource Management.

Future Scope:

- Integration with real-time recruitment portals.
- Resume parsing using Natural Language Processing (NLP).
- Deep learning models for improved prediction accuracy.

Address

Plot no 5, Krishna Tower, GopalNagar - A,
Gopalpura Bypass, Jaipur - 302015

Authorised Signatory

Phone

+91 93510 09002
0141 250 1609

Web/Email

www.linuxworldindia.org
lr@lwindia.com



Ref : LW/JPR/2026/1285

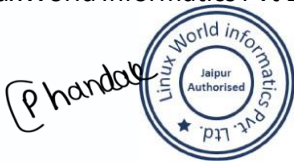
Date : 31st July 2026

- Web-based interface for recruiters.
- Automated interview scheduling.
- Candidate ranking system.
- Explainable AI (XAI) for prediction transparency.
- Cloud deployment using AWS or Azure.
- Integration with Applicant Tracking Systems (ATS).

We wish him all the success for his future

Regards

LinuxWorld Informatics Pvt Ltd



Authorised Signatory

● Address

Plot no 5, Krishna Tower, GopalNagar - A,
Gopalpura Bypass, Jaipur - 302015

● Phone

+91 93510 09002
0141 250 1609

● Web/Email

www.linuxworldindia.org
lr@lworldia.com